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ORIGINAL RESEARCH

STRUCTURAL

Transcatheter Edge-to-Edge Repair in Patients With Primary Tricuspid Regurgitation

Atsushi Sugiura, MD, PhD,[a] Julien Dreyfus, MD, PhD,[b] Sara Bombace, MD,[c] Maria Ivannikova, MD,[d] Joanna Bartkowiak, MD,[e] Stephan Haussig, MD,[f] Leonhard Moritz Schneider, MD,[g] Mohammad Kassar, MD,[h] Patrick Horn, MD,[i] Maurizio Taramasso, MD,[j] Christos Iliadis, MD,[k] Itsuki Osawa, MD,[l] Tadahiro Goto, MD, MPH, PhD,[m,n] Marcel Weber, MD,[a] Tetsu Tanaka, MD,[a] Sebastian Zimmer, MD,[a] Jean-François Obadia, MD, PhD,[o] Gilbert Habib, MD,[p] Baptiste Bazire, MD,[b] Bernard Iung, MD,[q] Yohan Bohbot, MD, PhD,[r] Christophe Tribouilloy, MD, PhD,[r] Erwan Donal, MD, PhD,[s] Mohammed Nejjari, MD,[b] Elisabeth Riant, MD,[b] Thierry Le Tourneau, MD, PhD,[t] Yoan Lavie-Badie, MD,[u] Augustin Coisne, MD, PhD,[v] Thomas Modine, MD, PhD,[w] Pascal Lim, MD, PhD,[x] Fabien Doguet, MD,[y] Christine Selton-Suty, MD,[z] Stephan Baldus, MD,[n] Malte Kelm, MD,[i] Fabien Praz, MD,[h] Wolfgang Rottbauer, MD,[g] Axel Hans-Peter Linke, MD,[f] Rebecca Hahn, MD,[e] Rudolph Volker, MD,[d] David Messika-Zeitoun, MD, PhD,[aa] Philip Lurz, MD, PhD,[c] Georg Nickenig, MD[a]

ABSTRACT

BACKGROUND Tricuspid regurgitation (TR) leads to right heart congestion and increased mortality risk. Unlike secondary TR, primary TR results from leaflet degeneration. Transcatheter edge-to-edge repair (TEER) is widely used for TR. However, its feasibility primary TR remains uncertain.

OBJECTIVES The aim of this study was to assess the safety and feasibility of TEER in patients with primary TR.

METHODS The primary TR registry is a multicenter cohort study of patients with primary TR undergoing TEER. Echocardiographic assessment classified leaflet pathology into type 1 (flail), type 2 (billowing prolapse), type 3 (perforation), and type 4 (restricted mobility). The primary endpoint was TR reduction to moderate or less at discharge.

RESULTS From December 2016 to April 2023, 114 patients (mean age 79.9 years, 53.5% men) were included. Most patients were in NYHA functional class III or IV (83.3%), with a median TRI-SCORE of 5.0. The TR pathologies were type 1 (28.1%), type 2 (61.4%), and type 4 (10.5%), with no type 3 cases. Device deployment was achieved in 95.6%, and 83.3% showed TR reduction to moderate or less. In-hospital mortality was 1.8%, and single-leaflet device attachment occurred in 3.5%. At 1 year, 79.7% of patients had TR moderate or less, with significant reductions in vena contracta (−5.0 mm; $P < 0.001$), annular diameter (−2.0 mm; $P = 0.003$), and mid right ventricular diameter (−3.0 mm; $P < 0.001$). NYHA functional class also improved significantly (NYHA functional class I or II: 17.1% at baseline vs 66.5% at follow-up; $P < 0.001$).

CONCLUSIONS TEER is a safe and effective option for primary TR, promoting right heart reverse remodeling and symptomatic relief, offering a vital alternative to surgery in selected patients. (JACC Cardiovasc Interv. 2025;18:1289-1299) © 2025 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

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ABBREVIATIONS AND ACRONYMS

RV = right ventricle/ventricular

RVFAC = right ventricular fractional area change

TAPSE = tricuspid annular plane systolic excursion

TEER = transcatheter edge-to-edge repair

TR = tricuspid regurgitation

Tricuspid regurgitation (TR) is a prevalent cardiovascular disorder, which is linked to limited functional physical capacity and an increased mortality risk. TR decreases cardiac output and worsens right heart congestion,[1] thereby leading to dyspnea, peripheral edema, and excessive fatigue. Most TR has a secondary etiology, whereas primary TR occurs in 5% to 20% of patients,[2] in whom leakage is due mainly to tricuspid valve leaflet degeneration. Surgery is considered the first-line treatment for primary TR;[3,4] however, in daily practice, many patients with TR are considered to have right ventricular (RV) dysfunction or end-organ damage at the time of referral for surgery. Persistent TR can lead to RV remodeling and comorbidities, posing those surgical risks. Surgical intervention for primary TR had a high reported operative mortality.

Innovations in transcatheter procedures have revolutionized the field of TR treatment. Transcatheter edge-to-edge repair (TEER) is a widely used technology that approximates leaflets using a clipping device, demonstrating the safety and effectiveness in reducing TR and alleviating heart failure symptoms after the reduction in TR.[5,6] However, the feasibility of TEER for primary TR remains unclear, because the previous studies focused mainly on secondary TR.[3,7] The main issue in primary TR is the pathologic tricuspid leaflets themselves, which may vary among individuals. Understanding whether TEER can efficiently treat patients with primary TR is crucial for clinical practice: there remains an unmet need for safe and effective alternatives for the treatment of primary TR. Also, correction of TR might lead to a reverse remodeling of the RV.

To address this gap, we conducted a multicenter study to evaluate the safety and feasibility of tricuspid TEER for primary TR. We examined TR reduction, postprocedural adverse events, and changes in RV parameters.

METHODS

STUDY SETTING AND POPULATION. The primary TR registry is an international, multicenter, retrospective data collection of patients with primary TR who underwent TEER with the MitraClip (Abbott Laboratories), TriClip (Abbott Laboratories), or PASCAL (Edwards Lifesciences) system from February 2016 to April 2023 across 3 countries (Germany, Switzerland, and the United States) and 10 centers. For statistical purposes, we integrated a multicenter French surgical data set of patients who underwent isolated tricuspid valve surgery for primary TR at 12 tertiary French centers from January 2007 to December 2017.[8] The institutional ethics committee of the University of Bonn approved the registry linked to the present study. The study was conducted in concordance with the Declaration of Helsinki. Patient data were

From the [a]Department of Internal Medicine II, University Hospital Bonn, Bonn, Germany; [b]Department of Cardiology, Centre Cardiologique du Nord, Saint-Denis, France; [c]Department of Cardiology, University Medical Center of Johannes Gutenberg University Mainz, Mainz, Germany; [d]General and Interventional Cardiology, Heart and Diabetes Center NRW, University Hospital of the Ruhr University Bochum, Bad Oeynhausen, Germany; [e]Cardiology Department, NewYork-Presbyterian/Columbia University Medical Center, New York, New York, USA; [f]Department for Cardiology, University Heart Center Dresden, Dresden, Germany; [g]Department of Cardiology, Ulm University Heart Center, Ulm, Germany; [h]Department of Cardiology, Inselspital, University of Bern, Bern, Switzerland; [i]Department of Cardiology, Pulmonology, and Vascular Diseases, University Hospital Düsseldorf, Düsseldorf, Germany; [j]HerzZentrum Hirslanden Zürich, Switzerland; [k]Department for Internal Medicine III, Faculty of Medicine and University Hospital Cologne, Cologne, Germany; [l]Division of Cardiology, Department of Medicine, Columbia University Irving Medical Center, New York, New York, USA; [m]Department of Clinical Epidemiology and Health Economics, School of Public Health, University of Tokyo, Tokyo, Japan; [n]TXP Medical, Tokyo, Japan; [o]Department of Cardiovascular Surgery and Transplantation, Louis Pradel Cardiovascular Hospital, Claude Bernard University, Bron, France; [p]Cardiology Department, APHM, La Timone Hospital, Marseille, France; [q]Cardiology Department, Bichat Hospital, Assistance Publique-Hôpitaux de Paris, and INSERM LVTS U1148, Université de Paris, Paris, France; [r]Department of Cardiology, Amiens University Hospital, Amiens, France; [s]Cardiology Department, LTSI UMR1099, INSERM, Université de Rennes-1, CHU de RENNES, Rennes, France; [t]CNRS, INSERM, l'Institut du Thorax, Université de Nantes, CHU de Nantes, Nantes, France; [u]Heart Valve Center, Toulouse University Hospital, Toulouse, France; [v]Department of Clinical Physiology and Echocardiography-Heart Valve Clinic, CHU Lille, Lille, France; [w]Department of Cardiology and Cardio-Vascular Surgery, Hôpital Cardiologique de Haut-Leveque, Bordeaux University Hospital, Bordeaux, France; [x]Cardiology Department, Expert Valve Center, Henri Mondor Hospital, Créteil, France; [y]Department of Cardiology and Cardio-Vascular Surgery, CHU Charles Nicolle, Rouen, France; [z]Cardiology Department, CHU Nancy-Brabois, Nancy, France; and the [aa]Division of Cardiology, University of Ottawa Heart Institute, Ottawa, Ontario, Canada.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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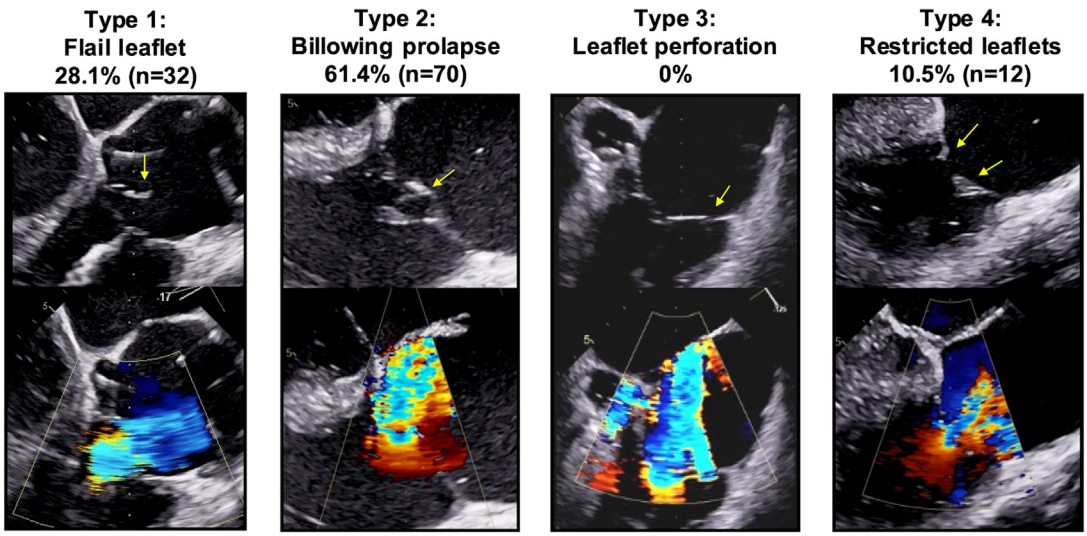
FIGURE 1 Pathologies of Primary Tricuspid Regurgitation



Shown are representative images of each pathologic feature of primary tricuspid regurgitation. The tricuspid valve pathologies included flail leaflet, billowing prolapse including Barlow's disease, leaflet perforation, and restricted leaflets in both systole and diastole due to leaflet degeneration.

collected at each local center and centralized in a unique anonymized database at University Hospital Bonn. All investigators were responsible for the collection and completion of data. Data analysis was performed at the University of Bonn. The study was approved by the Institutional Review Boards of the individual centers.

ECHOCARDIOGRAPHY ASSESSMENTS. Echocardiographic images of the tricuspid pathologies were assessed at each center. The definition of primary TR was a lack of leaflet coaptation due to intrinsic changes leading to restricted or excessive leaflet mobility or leaflet perforation. The predefined pathologies of primary TR were the following: type 1, flail leaflet; type 2, billowing prolapse including Barlow's disease; type 3, leaflet perforation; and type 4, restricted leaflets in both systole and diastole (**Figure 1**). A detailed definition of each pathology is provided in Supplemental Table 1.

Other echocardiographic assessments were analyzed according to the current guidelines.[9] The severity of TR was graded using 5 grades: 0+, none or trivial; 1+, mild; 2+, moderate; 3+, severe; 4+, massive; and 5+, torrential.[10] Other quantification parameters of TR included vena contracta width and effective regurgitant orifice area. Mid-RV diameter and the tricuspid annulus were measured at end-diastole. Tricuspid annular plane systolic excursion (TAPSE) and RV fractional area change (RVFAC) were assessed for RV function. We retrospectively assessed follow-up echocardiographic data obtained between 1 and 3 months and 12 months after TEER.

OUTCOME MEASURES. The primary endpoint of the study was a reduction in TR to moderate or less at discharge after tricuspid TEER. The secondary endpoints were the implantation success of TEER devices, in-hospital mortality, the duration of hospital stay, myocardial infarction, stroke, embolization of devices, and emergency surgery after TEER. Successful device deployment was defined as the successful delivery and deployment of the device systems with correct positioning of the intended devices without requiring implantation of additional devices. Additionally, event rates at 30 days were reported, and follow-up data were used to estimate 1-year mortality and heart failure rehospitalization rates. Clinical follow-up data were obtained through interviews at scheduled outpatient clinic visits or telephone interviews with their family.

For the multicenter French surgical data set, survival and heart failure hospitalization data after surgery were collected through hospital chart reviews

and telephone interviews with patients or family members, which were used to compare clinical outcomes with those after TEER.

STATISTICAL ANALYSIS. Categorized variables are presented as numbers and percentages. The normality of continuous measures was checked by visual examination of histograms and the Shapiro-Wilk test. Normally distributed variables are presented as mean ± SD and were compared using Student's *t*-test, while non-normally distributed variables are reported as median (Q1-Q3) and were compared using the Mann-Whitney *U* test. A paired Student's *t*-test was used to compare a series of continuous variables, whereas McNemar's test was used to compare paired categorical data.

We represented the distribution for the morphologic characteristics of primary TR. The rate of TR reduction to moderate or less at discharge is provided. The procedural outcome was also assessed according to predefined subcategories, including implanted device (MitraClip, TriClip, or PASCAL), leaflet morphology (type 1, type 2, type 3, type 4), and the experience of TEER for primary TR (first 5, 5th to 9th, and 10th or more). Furthermore, we assessed postinterventional changes in TR severity, echocardiographic parameters of the right heart, and NYHA functional class from baseline to 1-year follow-up.

To examine whether TEER can be an alternative to surgical treatment, we integrated the surgical cohort into the survival analysis. First, we examined the reductions in TR, in-hospital mortality, and hospital duration between the TEER and surgery groups. Second, to examine 1-year mortality between the TEER and surgical groups, we plotted time-to-event curves for all-cause mortality and heart failure rehospitalization up to 1 year after intervention using the Kaplan-Meier method. Third, we also compared the time-to-event risk for mortality between the TEER and surgical groups using Cox proportional hazards models adjusted for age, NYHA functional class, and TRI-SCORE, which were selected on the basis of earlier literature and clinical knowledge.[8,11,12] Fourth, we matched the patients of the TEER and surgery cohorts. In this matching process, we used the nearest neighbor algorithm with a caliper set of 0.10 for propensity scores, calculated by the following potential confounders: patient age, sex, diabetes mellitus, chronic obstructive pulmonary disease, atrial fibrillation, NYHA functional class, use of diuretic agents, European System for Cardiac Operative Risk Evaluation II, TRI-SCORE, TR severity, left ventricular ejection fraction, and TAPSE.[8,13,14] In this matched cohort, we repeated the survival analysis.

Additionally, we fitted a logistic regression analysis to examine the association of TR reduction with mortality, adjusted for treatment type, age, NYHA functional class, and TRI-SCORE.[12,15]

A sample size calculation was performed with a type 1 alpha error rate of 0.05 and a type 2 beta error rate of 0.2 (corresponding to power of 80%). The primary endpoint rate was tested against a prespecified performance goal of 80%, assuming a rate at discharge of 65%. On the basis of the calculation, the minimum required sample size was determined to be 44 patients.

All statistical analyses were performed using EZR version 1.37 (Saitama Medical Center, Jichi Medical University), R version 4.3.0 (R Foundation for Statistical Computing), or SPSS Statistics version 26 (IBM).

RESULTS

BASELINE CHARACTERISTICS. Among 114 patients included in the present analysis from 10 centers in Germany, Switzerland, and North America, 53.5% were men, the mean age was 79.9 years, the median European System for Cardiac Operative Risk Evaluation II was 5.2%, and the median TRI-SCORE was 5.0 points (**Table 1**). The study participants were highly symptomatic (83.3% were in NYHA functional class III or IV), and significant comorbidities included coronary artery disease (44.7%), atrial fibrillation (85.1%), and history of cardiac surgery (33.3%). At baseline, 21.9% of patients had incidental cardiac implantable electronic devices. The median N-terminal pro-B-type natriuretic peptide level was 1,694 ng/mL, and 90.4% of patients were taking diuretic agents at baseline.

ECHOCARDIOGRAPHIC CHARACTERISTICS. TR grade was severe or greater in 98.2%, with massive or torrential TR in 64.0%. The mean mid-RV diameter was 36.8 ± 7.5 mm, the mean tricuspid annular diameter 43.5 ± 7.1 mm, and the mean estimated pulmonary artery systolic pressure 43.4 ± 13.7 mm Hg (**Table 2**). The tricuspid valve pathologies (**Figure 1**) included flail leaflet (type 1, 28.1% [n = 32]), billowing prolapse (type 2, 61.4% [n = 70]), and restricted leaflets in both systole and diastole (type 4, 10.5% [n = 12]). In contrast, no leaflet perforation was observed in this cohort (type 3, 0%). Among 102 patients with leaflet prolapse (ie, type 1 and type 2), 52% showed a single pathologic segment, whereas 46% had multiple segments of prolapse (Supplemental Figure 1). The mean height of the prolapse was 5.8 ± 3.5 mm.

TABLE 1 Baseline clinical Characteristics (N = 114)

Age, y	79.9 ± 7.2
Male	61 (53.5)
Body mass index, kg/m^2	25.0 ± 4.4
EuroSCORE II, %	5.2 (3.1-8.8)
TRI-SCORE, points	5.0 (4.0-6.0)
Diabetes mellitus	21 (18.4)
Arterial hypertension	97 (85.1)
Chronic obstructive pulmonary disease	21 (18.4)
Coronary artery disease	51 (44.7)
Stroke	16 (14.0)
Peripheral artery disease	20 (17.5)
Prior coronary artery bypass graft	21 (18.4)
Prior valvular surgery	25 (21.9)
Atrial fibrillation	100 (87.7)
NYHA functional class	
II	17 (14.9)
III	90 (78.9)
IV	5 (4.4)
CIED	25 (21.9)
Total bilirubin, mg/dL	0.86 ± 0.48
Hemoglobin, g/dL	12.1 ± 2.3
eGFR, mL/min/1.73 m^2	58.9 ± 20.6
NT-proBNP, ng/mL	1,694 (1,296-3,400)
Diuretic agent	103 (90.4)
Anticoagulant	
Warfarin	23 (20.2)
DOAC	67 (58.8)
Beta-blocker	88 (77.2)
ARB/ACE-I/ARNI	71 (62.3)
MRA	40 (35.1)

Values are mean ± SD, n (%), or median (Q1-Q3).

ACE-I = angiotensin-converting enzyme inhibitor; ARB = angiotensin receptor blocker; ARNI = angiotensin receptor neprilysin inhibitor; CIED = cardiac implantable electronic device; DOAC = direct oral anticoagulant agent; eGFR = estimated glomerular filtration rate; EuroSCORE = European System for Cardiac Operative Risk Evaluation; MRA = mineralocorticoid receptor antagonist; NT-proBNP = N-terminal pro-B-type natriuretic peptide.

TABLE 2 Baseline Echocardiographic Parameters (N = 114)

LV ejection fraction, %	56.4 ± 10.0
LV end-diastolic volume, mL	86.6 ± 39.8
Left atrial volume, mL	90.9 ± 39.6
TR severity	
Moderate	2 (1.8)
Severe	39 (34.2)
Massive	43 (37.7)
Torrential	30 (26.3)
TR vena contracta, mm	10.9 ± 4.0
TR EROA, mm^2	71.9 ± 39.6
Tricuspid regurgitant volume, mL	60.7 ± 27.6
Tricuspid annulus diameter, mm^2	43.5 ± 7.1
Mid-RV diameter, mm^2	36.8 ± 7.5
RV end-diastolic area, cm^2	24.4 ± 7.7
RV end-systolic area, cm^2	14.0 ± 5.5
Right atrial area, cm^2	31.2 ± 12.4
RVFAC, %	43.6 ± 10.1
TAPSE, mm	17.8 ± 6.0
TR pressure gradient, mm Hg	34.0 ± 13.9
IVC diameters, mm	23.7 ± 5.5
Estimated pulmonary artery pressure, mm Hg	43.4 ± 13.7

Values are mean ± SD or n (%).

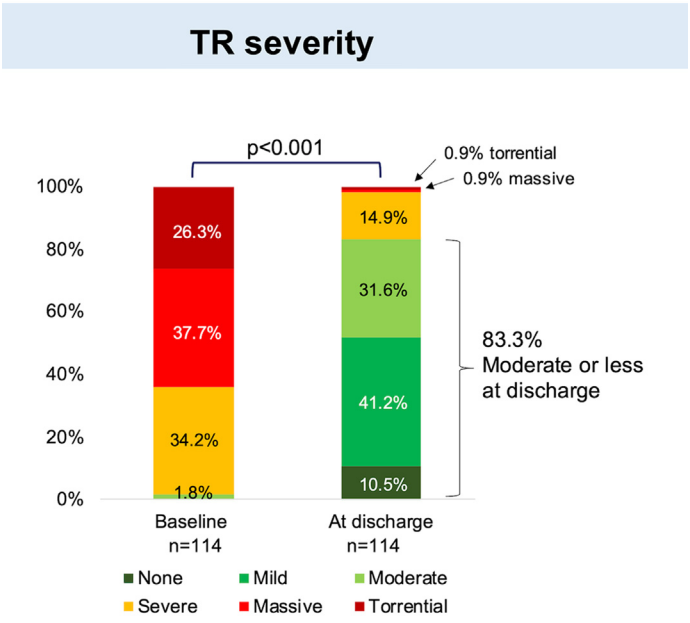
EROA = effective regurgitant orifice area; IVC = inferior vena cava; LV = left ventricular; RV = right ventricular; RVFAC = right ventricular fractional area change; TAPSE = tricuspid annular plane systolic excursion; TR = tricuspid regurgitation.

PROCEDURAL OUTCOMES. There was a significant reduction in the severity of TR. TR reduction to moderate or less at discharge was achieved in 95 patients (83.3%) (**Figure 2**). The MitraClip, TriClip, and PASCAL systems were used in 26.3%, 39.5%, and 34.2% of patients, respectively (**Table 3**). One device was implanted in 42 patients (36.8%), 2 devices in 54 (47.4%), and ≥3 devices in 17 (14.9%), with an average number of devices of 1.8 ± 0.7 and a mean device time of 107.3 ± 48.6 minutes. The devices were placed on the anteroseptal leaflets in 148 (75.1%), the posteroseptal leaflets in 46 (23.4%), and the anteroposterior leaflets in 3 (1.5%) (Supplemental Figure 2). Successful device deployment was achieved in 109 of the 114 study participants (95.6%). There was no periprocedural death, myocardial infarction, stroke, embolization of devices, or emergency surgery after TEER. The in-hospital mortality rate was 1.8% (n = 2 of 114), and no further deaths were observed up to 30-day follow-up, resulting in a 30-day mortality rate of 1.8%. Single-leaflet device attachment occurred in 4 patients (3.5%). Thirty-day outcomes are listed in Supplemental Table 2.

TR reduction according to the predefined subgroups is summarized in Supplemental Figure 3. The baseline characteristics and detailed procedural outcomes are provided in Supplemental Tables 3 and 4. TR reduction was consistently observed among the MitraClip and TriClip (81.3%) and PASCAL (87.2%) systems. For the subgroup of pathologic types, the baseline characteristics are listed in Supplemental Tables 5 and 6. Type 1 and type 4 pathologies were characterized as showing more prominent TR at baseline. After TEER, reduction in TR to moderate or less was observed in 81.2% for type 1, 87.1% for type 2, and 66.7% for type 4. Furthermore, procedural outcomes were assessed according to the experience of TEER for primary TR. There was a trend toward an increase in the rate of TR reduction to moderate or less along with the number of procedural experiences: 74.5% for the first 5 cases, 84.8% for the 6th to 10th cases, and 94.1% for the 11th and subsequent cases.

JACC: CARDIOVASCULAR INTERVENTIONS VOL. 18, NO. 10, 2025
MAY 26, 2025:1289-1299

FIGURE 2 Reduction in TR After Transcatheter Edge-to-Edge Repair



Tricuspid regurgitation (TR) severity was significantly reduced from baseline to at discharge, with 83.3% of patients showing TR reductions to moderate or less.

TR SEVERITY AND RIGHT HEART PARAMETERS DURING FOLLOW-UP. At 1 year, a paired analysis of TR grades was available in 94 patients. In the analysis (median 163 days; Q1-Q3: 80-363 days), 79.7% of patients had TR moderate or less during follow-up (Figure 3). Reduction in TR of at least 1 grade was observed in 91.4% of patients, while 71.3% of patients showed TR reductions of ≥2 grades. Also, TR vena contracta was significantly reduced (median −5.0 mm; Q1-Q3: −8.1 to −2.2 mm). Changes from baseline to follow-up echocardiography are summarized in Table 4. There were reductions in tricuspid annular diameter (median −2.0 mm; Q1-Q3: −6.0 to 1.8 mm), mid-RV diameter (median −3.0 mm; Q1-Q3: −8.5 to 1.0 mm), and RVFAC (median −5.7%; Q1-Q3: −10.4% to 2.6%). There was an insignificant decrease in TAPSE during the follow-up (Figure 4).

TABLE 3 Procedural Outcomes (N = 114)

Implantation success	109 (95.6)
Device time, min	107.3 ± 48.6
Mean number of implanted devices	1.8 ± 0.7
Type of TEER devices	
MitraClip	30 (26.3)
TriClip	45 (39.5)
PASCAL	39 (34.2)
In-hospital mortality	2 (1.8)
Myocardial infarction	0
Stroke	0
Device embolization	0
Emergency surgery	0
Single leaflet device attachment	4 (3.5)
Vascular complications	4 (3.5)

Values are n (%) or mean ± SD.
TEER = transcatheter edge-to-edge repair.

1-YEAR CLINICAL OUTCOMES. Of 114 patients in the whole cohort, 94 patients completed 1-year clinical follow-up for prespecified clinical outcomes. Two patients underwent transcatheter reintervention within 3 months: 1 patient for recurrent severe TR and the other for late occurring single-leaflet device attachment. With median follow-up of 230 days (Q1-Q3: 90-370 days), 12 of 114 patients died, and 9 experienced heart failure rehospitalization. The estimated 1-year mortality was 17.3% (95% CI: 10.0%-29.0%), while the estimated event rate of heart failure rehospitalization was 11.1% (95% CI: 3.7%-17.9%).

We integrated the multicenter surgical database into the analysis (Supplemental Table 7). Median follow-up duration after surgery was 925 days (Q1-Q3: 178-1,789 days). Compared with the TEER group, TR reduction was more pronounced in the surgical cohort (TR reduction to moderate or less at discharge: 83.3% vs 97.3%; $P < 0.001$). In contrast, in-hospital mortality tended to be lower in the TEER group compared with the surgical group (1.8% vs 6.3%; $P = 0.062$). There was a significant difference in length of hospital stay (3 days [Q1-Q3: 2-4 days] vs 13 days [Q1-Q3: 9-21 days]; $P < 0.001$). In the Kaplan-Meier analysis, the outcomes were comparable between the TEER and surgical cohorts for mortality (17.3% vs 9.0%; log-rank $P = 0.25$) and for the composite of mortality and heart failure rehospitalization (25.7% vs 21.6%; log-rank $P = 0.73$). In the multivariable Cox proportional hazards model, TEER appeared to be comparable with surgery in terms of survival (HR: 0.50; 95% CI: 0.20-1.25; $P = 0.14$) and the composite outcomes (HR: 0.70; 95% CI: 0.34-1.43; $P = 0.33$). Similarly, there were no differences in outcomes in the propensity score-matched cohort (mortality 10.1% vs 15.2% [log-rank $P = 0.74$]; composite of mortality and hospitalization 15.0% vs 26.5% [log-rank $P = 0.68$]) (Supplemental Table 8).

As for the association between TR reduction and 1-year outcomes, TR reduction to moderate or less at discharge was associated with lower risk for mortality (adjusted OR: 0.25; 95% CI: 0.07-0.96).

In terms of heart failure symptoms in patients undergoing TEER, there was a significant

improvement in NYHA functional class: the percentage of patients in NYHA functional class I to II increased from 17.1% at baseline to 66.5% at follow-up ($P < 0.001$) (Supplemental Figure 4).

DISCUSSION

In this multicenter cohort study (**Central Illustration**), we found that the tricuspid TEER for treating primary TR was safe and feasible: the rate of TR reduction to moderate or less was 83.3%, and in-hospital mortality was 1.8%. The reduction in TR was sustainable during 1-year follow-up. Furthermore, there were signs of reverse remodeling of the right heart and alleviation of heart failure symptoms.

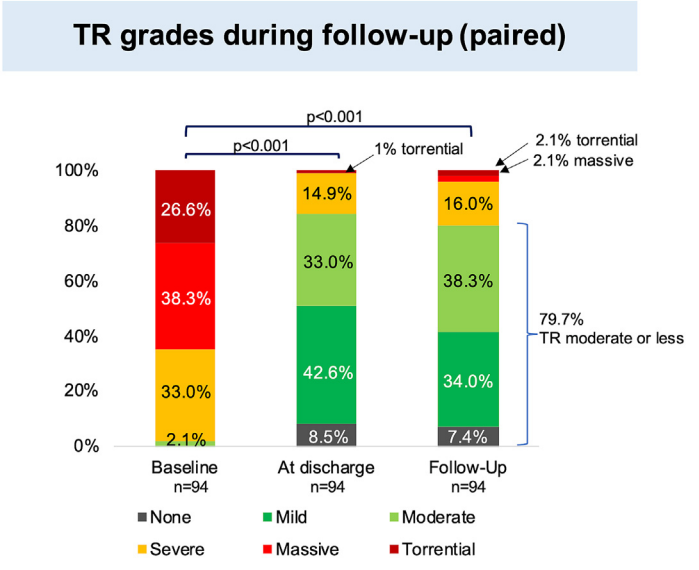
Various leaflet morphologies of primary TR were reported in these multicenter registry data. Flail leaflet was observed in 28.1%, billowing prolapses in 61.4%, and shortened or thickened leaflets in 10.5%. No patients were treated with the tricuspid TEER for a leaflet perforation. Additionally, the tricuspid annulus and RV were also modestly dilated. The study participants were characterized as being at old age, highly symptomatic, and at high risk for surgery.

Under the complex conditions of the tricuspid leaflets and multiple comorbidities, successful device implantation was achieved in 95.5%. There was no periprocedural death or emergency surgery after TEER, highlighting the safety of this procedure. The procedural outcomes were comparable with the data from the bRIGHT (An Observational Real-World Study Evaluating Severe Tricuspid Regurgitation Patients Treated With the Abbott TriClip™ Device) study or the TriCLASP (Transcatheter Repair of Tricuspid Regurgitation With Edwards PASCAL Transcatheter Valve Repair System) trial.[5,16] TR was effectively reduced in the majority of patients. Device iterations may be attributable to the safety and effectiveness profile of the procedure: the larger devices (ie, MitraClip XT and XTR, TriClip XT and XTW, and PASCAL) were used in 90.4%, and the contemporary devices with optional independent grasping features (TriClip G4 and PASCAL) were used in 60.5% of the cohort.

The procedural outcomes are also likely to be related to operators' and imagers' experience. The rate of the TR reduction to moderate or less at discharge appeared to increase according to the degree of experience treating primary TR, suggesting a learning curve of TEER for primary TR. Also, a higher patient volume in the participating centers may be attributable to the observed findings. The percentage of 93.8% for achieving TR moderate or less was higher compared with the overall procedural outcome of TRILUMINATE early feasible study (57%) and comparable with the latest reports of tricuspid TEER (TRILUMINATE pivotal study, 87%; TriCLASP, 90%; bRIGHT, 80%).[5-7,16]

Procedural screening and patient selection are key to achieve an optimal procedural result.[2] Shortened leaflets may challenge tricuspid TEER to adequately reduce TR,[17] in which transcatheter tricuspid valve replacement technologies might be another viable option.[2,18,19] By contrast, flail leaflets preclude the device feasibility of transcatheter tricuspid valve

FIGURE 3 Severity in TR at Baseline and Follow-Up



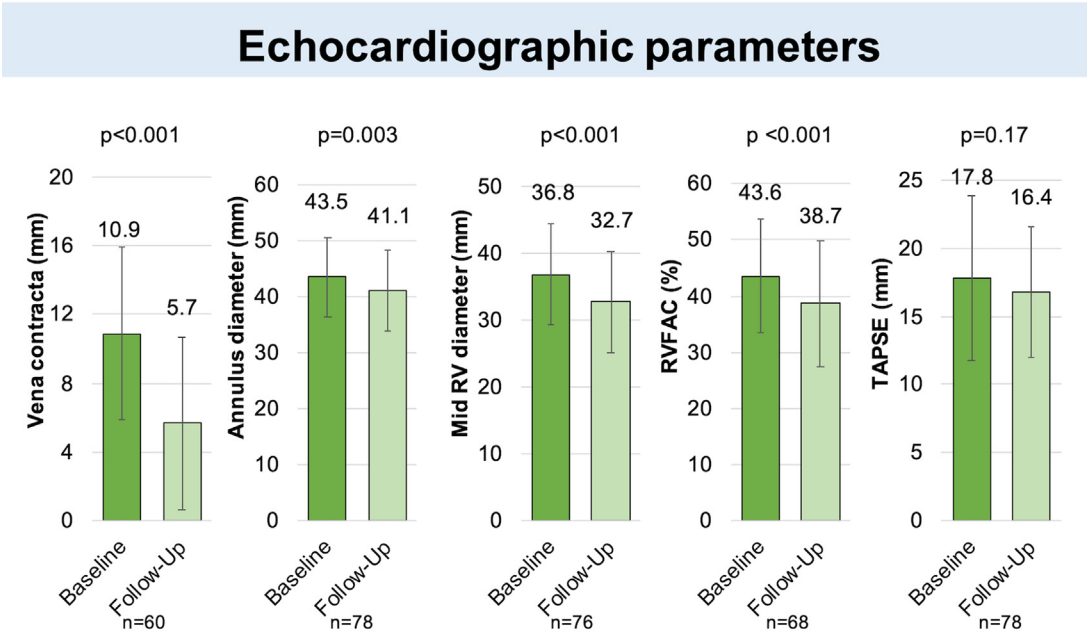
The paired analysis showed a durable reduction in tricuspid regurgitation (TR) during the follow-up. FU = follow-up.

TABLE 4 Paired Analysis of Echocardiographic Parameters at Baseline and Latest Available Follow-Up

	Baseline	Follow-Up	*P* Value
LVEF, %	56.1 ± 9.9 (79)	56.1 ± 8.5 (79)	0.51
Tricuspid annular diameter, mm	43.5 ± 7.1 (78)	41.1 ± 7.3 (78)	0.003
Mid-RV diameter, mm	36.8 ± 7.5 (76)	32.7 ± 7.5 (76)	<0.001
RV end-diastolic area, mm^2	24.4 ± 7.7 (68)	22.4 ± 8.1 (68)	0.031
RV end-systolic area, mm^2	14.0 ± 5.5 (68)	13.8 ± 6.0 (68)	0.52
RVFAC, %	43.6 ± 10.1 (68)	38.7 ± 11.1 (68)	<0.001
TAPSE, mm	17.8 ± 6.0 (78)	16.8 ± 4.8 (78)	0.17
TR vena contracta, mm	10.9 ± 4.0 (60)	5.7 ± 4.9 (60)	<0.001
TR pressure gradient, mm Hg	34.0 ± 13.9 (74)	34.0 ± 12.7 (74)	0.84

Values are the mean ± SD (number of patients for the paired comparison for each variable).
LVEF = left ventricular ejection fraction; other abbreviations in Tables 2 and 3.

FIGURE 4 Changes in Echocardiographic Findings From Baseline to Follow-Up



During follow-up, there were significant reductions in the vena contracta, tricuspid annular diameter, right ventricular (RV) diameter, and RV fractional area change (RVFAC). TAPSE = tricuspid annular plane systolic excursion.

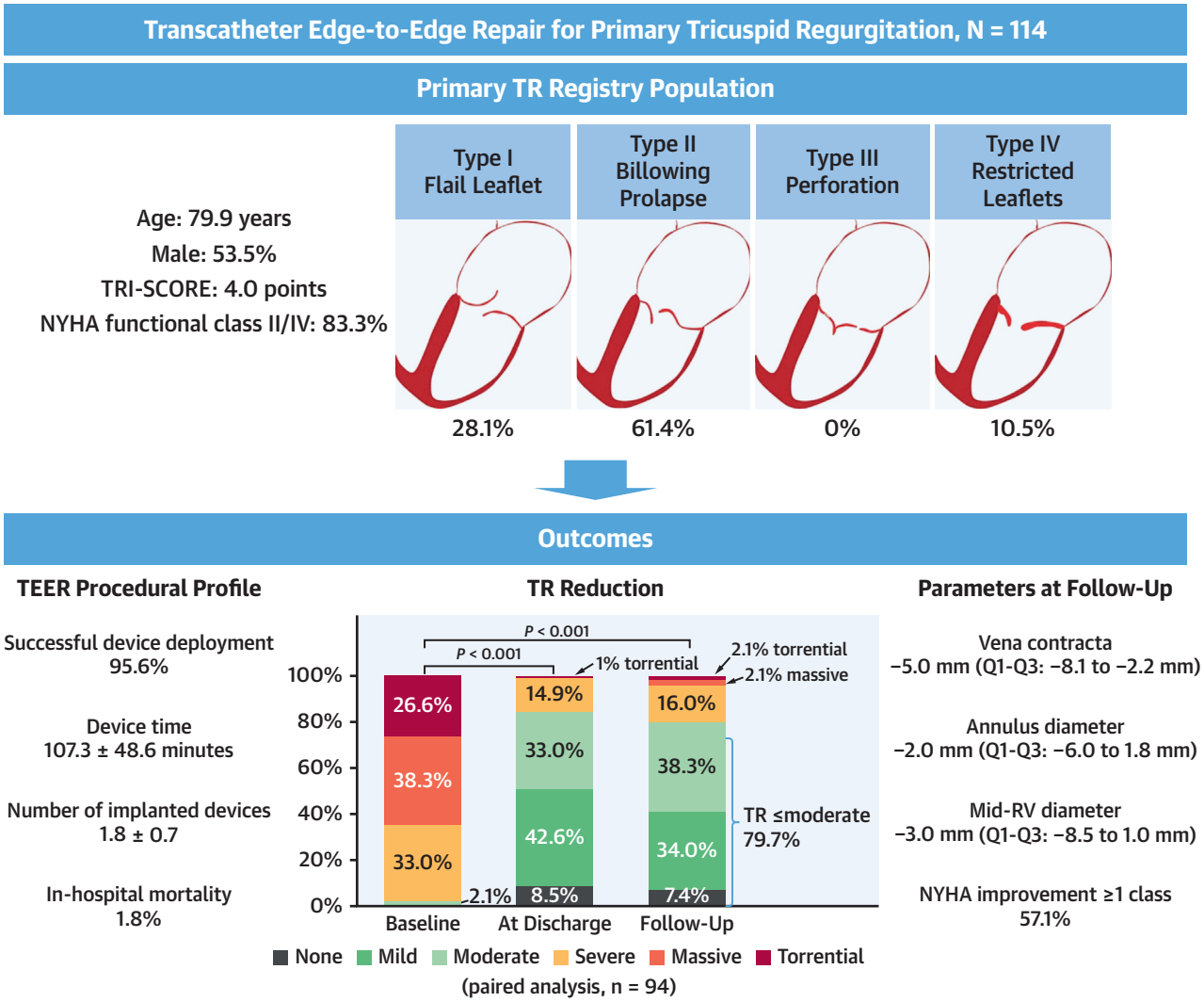
replacement.[20] The present analysis demonstrated that a flail leaflet or billowing prolapse, including Barlow's disease, can be well treated with TEER devices. In contrast, TEER for type 4 pathology might be suboptimal: we found that only 66% of these patients achieved TR reduction to moderate or less after TEER.

We demonstrated that the TR reduction after TEER was durable up to 1-year follow-up, and there was reverse remodeling of the RV after the treatment in patients with primary TR. The reductions in right heart dimension are likely a result of TR correction and subsequent decrease in RV volume.[21,22] Also, there were decent declines in RVFAC and TAPSE, which is in line with previous investigations for RV function after TR treatment.[22] This decline does not indicate a worsening of RV function. It may instead indicate a course of the TR reduction and a significant decrease in RV preload.[23]

The estimated 1-year mortality rate was 17.3%, comparable with real-world data on transcatheter tricuspid valve interventions (23%) but slightly higher compared with the data from the early feasibility studies (TRILUMINATE, 7.1%; CLASP TR [Edwards PASCAL Transcatheter Valve Repair System in Tricuspid Regurgitation], 10.8%).[24,25] The wide range of mortality among the studies might suggest that baseline comorbidities are delineating factors for the prognosis.[8,11] Noticeably, the study patients showed a significant improvement in heart failure symptoms as assessed using the NYHA functional classification.

Moreover, 1-year survival data suggest that clinical outcomes following TEER could be equivalent to those after surgery in selected patients with primary TR. The TEER approach was linked to lower in-hospital mortality compared with surgery, whereas surgery was superior in its ability to correct TR. Tricuspid TEER can reduce TR but rarely eliminate it. Primary TR is a very heterogeneous tricuspid apparatus disease, requiring heart team assessment, including a cardiac surgeon, an interventional cardiologist, and an echocardiographer.[2,26] Despite the comparable 1-year mortality after surgery and TEER in the present analysis, the surgical group patients may not have been anatomically eligible for TEER. In fact, in our cohort, there were no patients with leaflet perforation, and patients with restrictive leaflets tended to have suboptimal TR reduction after TEER. In contrast, our findings may indicate that TEER is a safe and effective alternative to surgery, if the aforementioned anatomies are precluded. Future investigations are needed to validate our preliminary results.

CENTRAL ILLUSTRATION **TEER for Treating Patients With Primary TR**



Sugiura A, et al. JACC Cardiovasc Interv. 2025;18(10):1289–1299.

This international multicenter registry demonstrated that transcatheter edge-to-edge repair (TEER) can be a safe, effective, and feasible option for treating primary tricuspid regurgitation (TR) in selected patients. Reduction in TR to moderate or less was achieved in 83.3%. The study participants experienced significant reverse remodeling of the right heart and marked reduction of heart failure symptoms.

STUDY LIMITATIONS. First, this study was a retrospective cohort study. Despite the international multicenter data collection, the study cohort and findings are subject to selection bias. Although each center in the primary TR registry also offers surgical options for patients with primary TR, the therapeutic decision making was discrete to each heart team, which might bias the study population and procedural findings of TEER.

Second, the sample size was relatively small, although this is the largest data set to explore patients with primary TR undergoing TEER.

JACC: CARDIOVASCULAR INTERVENTIONS VOL. 18, NO. 10, 2025
MAY 26, 2025:1289–1299

Furthermore, the study was designed to focus on the procedural safety and feasibility of TEER, but the follow-up rate was limited. Larger scale studies with longer follow-up data are mandatory to confirm the durability of the procedures.

Third, although we assessed and defined the morphologic characteristics of the tricuspid valve according to echocardiographic images, the underlying causes for pathologies, such as postbiopsy, post-trauma, and postendocarditis, were not investigated, because of the retrospective design. For instance, leaflet perforation due to endocarditis is one of the most prevalent features of primary TR in daily practice, but there were no patients with leaflet perforation in the present study cohort.

Fourth, we showed a significant reduction in TR accompanied by right heart reverse remodeling and improvement in NYHA functional class during the follow-up. However, whether treating primary TR with TEER reduces symptoms and improves survival is yet to be investigated in further studies and, ideally, in randomized controlled trials.

Last, it is essential to compare survival rates between surgery and TEER to estimate the potential of TEER as an alternative in this population. However, the surgical cohort differed greatly from TEER patients. Although we conducted a propensity score analysis accounting for baseline characteristics, a randomized control trial is needed to establish therapeutic strategies of surgery and TEER for these populations.

CONCLUSIONS

Transcatheter tricuspid edge-to-edge repair was safe and effective in treating primary TR in patients who are at high risk for surgery. The 1-year data on TEER showed durable TR reduction and reverse remodeling of the RV after TEER in patients with primary TR. Furthermore, 1-year clinical outcomes after TEER were comparable with surgical data, suggesting that TEER for primary TR can be a feasible alternative to surgical therapy in selected patients if anatomically amenable to TEER and at high surgical risk.

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This study received a research grant from Edwards Lifesciences. Dr Sugiura has received lecture and proctoring fees from Abbott and Edwards Lifesciences. Dr Dreyfus has received speaker and proctoring fees from Abbott. Dr Tanaka has received grants from the Japanese College of Cardiology and the Uehara Memorial Foundation; and has received honoraria from Canon Medical Systems. Dr Weber has received lecture and proctoring fees from Abbott and Edwards Lifesciences. Dr Messika-Zeitoun has received a grant from Edwards Lifesciences. Dr Nickenig has received research funding from Deutsche Forschungsgemeinschaft, the German Federal Ministry of Education and Research, the European Union, Abbott, Edwards Lifesciences, Medtronic, and St. Jude Medical; and has received honoraria for lectures and advisory board membership from Abbott, Edwards Lifesciences, Medtronic, and St. Jude Medical. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Atsushi Sugiura, Heart Center Bonn, Department of Medicine II, University Hospital Bonn, Venusberg-Campus 1, 53127 Bonn, Germany. E-mail: atsushi.sugiura@ukbonn.de.

PERSPECTIVES

WHAT IS KNOWN? TR leads to progressive right heart congestion, often causing right heart failure symptoms and increased mortality risk. Unlike secondary TR, primary TR is caused by degeneration of the tricuspid valve leaflets.

WHAT IS NEW? Transcatheter tricuspid edge-to-edge repair is safe and effective for reducing TR in patients with primary tricuspid valve disease. It leads to reverse remodeling of the right heart and symptomatic reduction over time.

WHAT IS NEXT? TEER for treating primary TR may be a viable alternative to surgery in selected patients if anatomically amenable to repair and at high surgical risk. The outcome benefit should be tested in randomized control trials.

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KEY WORDS primary tricuspid regurgitation, surgery, transcatheter edge-to-edge repair, tricuspid regurgitation

APPENDIX For supplemental tables and figures, please see the online version of this paper.